Deep MLP on FashionMNIST

Candidate number: `63439`

1 Introduction and data description

Introduction

This report studies a family of deep multi-layer perceptron (MLP) classifiers for 10-class image classification. Starting from a baseline MLP trained with cross-entropy loss on logits, we evaluate motivated variants to improve generalisation, including regularisation via dropout and weight decay, and an ablation using label smoothing (with mild training-time augmentation). Hyperparameters are tuned via targeted sweeps on a fixed held-out validation split. We monitor learning curves (training and validation loss over epochs) to diagnose overfitting, and apply early stopping in the sense of selecting the best epoch on the validation set, consistently across all runs. Validation accuracy (equivalently, validation error $1 - \text{accuracy}$) is the primary performance metric, and the final model is selected by validation performance.

Dataset

We use the FashionMNIST dataset introduced by Xiao et al. (2017). FashionMNIST contains 70,000 28×28 grayscale images of fashion products from 10 balanced categories (7,000 images per category). The dataset is constructed from Zalando product images: photographs are converted into a standardised 28×28 grayscale format through a fixed preprocessing pipeline (including resizing and centring on a canvas), and labels correspond to product silhouette categories curated by domain experts (Xiao et al., 2017; Zalando Research, 2017).

The dataset follows the same file format and split conventions as MNIST: 60,000 images are provided for training and 10,000 images are held out as the official test set (Xiao et al., 2017). Since no validation set is provided, we create a validation subset by splitting the training set once using a seeded permutation of indices; this split is held fixed across all hyperparameter sweeps.

Benchmark context. FashionMNIST was designed as a more challenging drop-in alternative to MNIST (Xiao et al., 2017). In Xiao et al. (2017), an MLP-type neural network baseline achieves test accuracy in the mid–high 0.8 range, which provides a useful reference point for interpreting deep MLP performance in our experiments.

2 Models and methods

Preprocessing and split

All images are converted to tensors and flattened to 784-dimensional vectors before entering the MLP. Since MLPs are permutation invariant, they treat each pixel as an independent feature, losing the spatial structure that a CNN would exploit. To ensure a robust evaluation of generalisation,

we create a fixed train/validation split from the original 60,000 training images using a seeded permutation. (`torch.randperm`)

Baseline architecture

Our baseline is a fully-connected deep MLP consisting of three hidden layers with widths of **512, 256, and 128** units, followed by a linear 10-way output layer. We use **ReLU** (Rectified Linear Unit) activations after each hidden layer to mitigate the vanishing gradient problem typically associated with deeper networks. This architecture is designed to be sufficiently expressive to "solve" the training set, allowing us to focus on the generalization gap rather than a lack of model capacity.

Loss function and outputs (logits + cross-entropy)

The model outputs *logits* (no softmax in the forward pass). Training minimises multi-class cross-entropy using `CrossEntropyLoss`, which applies a softmax internally and computes the negative log-likelihood of the correct class:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log p_\theta(y_i \mid \mathbf{x}_i)\,, \qquad p_\theta(y \mid \mathbf{x}) = \frac{\exp(z_y(\mathbf{x};\theta))}{\sum_{k=1}^{10}\exp(z_k(\mathbf{x};\theta))}.$$

Cross-entropy is appropriate for multi-class classification because it places an emphasis on penalising confident misclassifications.

The model optimises by using mini-batch training and monitoring the validation loss as the primary signal for early stopping. Specifically, the checkpoint that achieves peak validation accuracy within the epoch budget is reported, ensuring we capture the model before it begins to overfit the specific noise of the training data.

Variants: Regularisation and Tuning Strategy

To address the performance gap between training and validation sets, we implemented a sequential tuning strategy focused on improving model generalization. We first introduced dropout after each hidden ReLU activation layer. This technique prevents brittle feature co-adaptations by randomly zeroing activations during training, effectively forcing the network to learn more robust, redundant representations. With the optimal dropout rate established, we then tuned L_2 weight decay as a complementary control on model complexity. By penalising large weight values, weight decay encourages a smoother mapping and reduces the model's sensitivity to high frequency noise in the training data.

Once the regularization regime was stabilized, the learning rate was tuned to optimize the training dynamics under these new constraints. We also performed ablations using Batch Normalization and a combination of image augmentation with label smoothing. Batch Normalization was inserted before the ReLU activations to stabilize internal covariate shift, while label smoothing was tested to prevent the model from becoming overconfident in its predictions.

3 Results

Learning curves and overfitting behaviour

Figure 1 compares training and validation cross-entropy loss for the baseline and the selected final model. Both models show a steep early drop in loss (roughly epochs 1–3), indicating that the

network quickly learns coarse class structure. The separation emerges in mid training: the **baseline validation loss largely plateaus from around epochs $\approx$ 7–10 onward**, while baseline training loss continues to fall steadily through later epochs. This widening train–validation gap is consistent with overfitting: additional fitting of the training set is not translating into improved held-out performance.

The selected final model maintains a **lower validation loss throughout most of training** and continues to make gradual validation improvements further into training (before flattening late in the run). It is also typical that the regularised model can exhibit higher training loss than the baseline at comparable epochs, because regularisation (especially dropout) makes the effective training objective noisier/harder while improving generalisation on validation data.

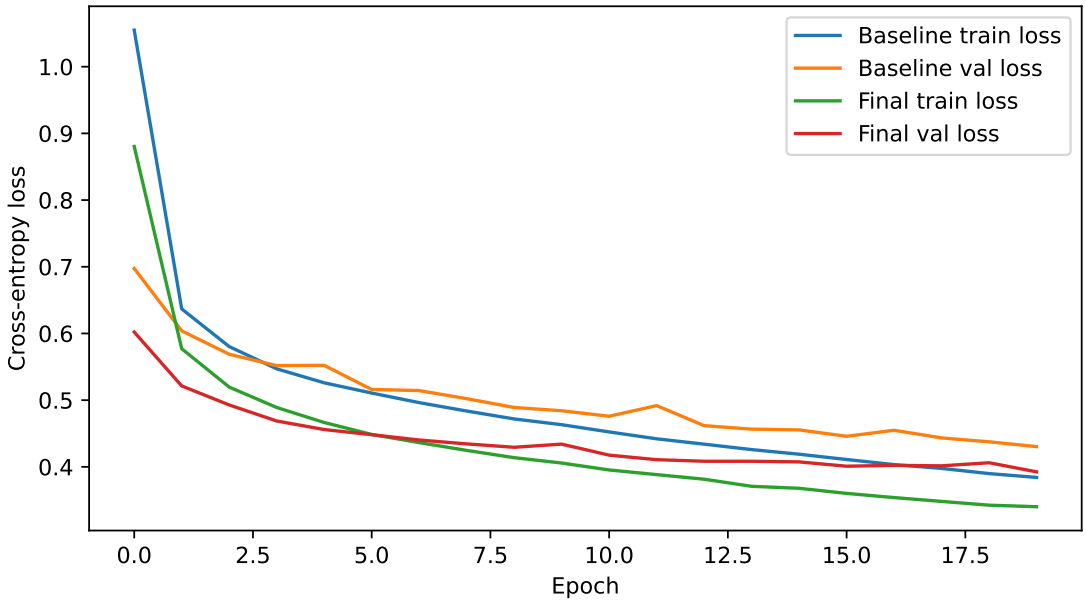

Figure 1: Training and validation cross-entropy loss over epochs for the baseline model and the selected final model.

Selection metric and early stopping

We select models by **peak validation accuracy** within each run (early stopping as model selection) and report the corresponding epoch (Table 1). Accuracy is a natural choice here because FashionMNIST is a balanced 10-class classification task: each example contributes equally to performance, and accuracy directly measures the proportion of correctly classified images. We also report **validation error** as $1 -$ accuracy, which is the misclassification rate.

Ablations, sequential tuning rationale, and final model

Table 1 summarises targeted sweeps and a sequential tuning strategy motivated by the learning-curve diagnosis. Since the baseline exhibits overfitting (Figure 1), we first introduce **dropout** and tune its rate to directly reduce reliance on brittle feature co-adaptations and narrow the train–validation gap. With dropout fixed at its best value, we then tune **weight decay** as a complementary control on model complexity: weight decay penalises large weights and tends to smooth the learned mapping, which is most informative to test once co-adaptation is already reduced. Finally, after the

Table 1: Model selection / ablation results on the fixed validation split. Validation error is $1 - \text{acc}$. Best epoch is the epoch of peak validation accuracy within each run.

Run	Dropout	WD	LR	Peak acc	Val err	Best ep.
Baseline	0.0	0	1×10^{-4}	0.8835	0.1165	19
Best dropout	0.2	0	1×10^{-4}	0.8898	0.1112	20
Best weight decay	0.2	1×10^{-4}	1×10^{-4}	0.8894	0.1106	19
Final selected	**0.2**	1×10^{-4}	3×10^{-4}	**0.8982**	**0.1018**	**18**
BatchNorm attempt	0.2	1×10^{-4}	3×10^{-4}	0.8972	0.1028	17
Augment + label smoothing[†]	–	1×10^{-4}	1×10^{-4}	0.8874	0.1126	40

[†]Accuracy evaluated sparsely (epoch 1, every 5 epochs, and final epoch) and using a different training pipeline; values are best observed under that schedule, so this row is less directly comparable to the per-epoch sweeps above.

regularisation regime is set, we tune the **learning rate** because it primarily governs optimisation dynamics (how efficiently the optimiser reaches a good solution under the regularised objective) rather than the effective capacity of the model.

The final selected configuration—**dropout** $p = 0.2$ **+ weight decay** 10^{-4} **with LR** 3×10^{-4}—achieves peak validation accuracy **0.8982** at epoch 18, corresponding to validation error **0.1018**. Relative to the baseline (0.8835; error 0.1165), this is an absolute reduction in validation error of **1.47 percentage points**. Empirically, most of the gain comes from introducing dropout, weight decay provides a smaller additional improvement, and learning-rate tuning produces the final uplift.

Alternative ablations

BatchNorm is competitive (0.8972) but does not exceed the selected configuration, suggesting limited additional generalisation benefit beyond dropout + weight decay in this MLP setting. The augmentation + label smoothing attempt underperforms relative to the best regularised baseline; with a pixel-flattened MLP, image-space augmentation may be less naturally exploited, and label smoothing can reduce the effective target confidence and lead to mild underfitting unless other hyperparameters are re-tuned.

4 Conclusion

This report benchmarks a deep MLP for FashionMNIST and shows that performance is primarily limited by overfitting rather than a lack of capacity. Learning curves reveal that the baseline model's validation loss plateaus mid-training while training loss continues to decrease, motivating regularisation-focused improvements. A targeted sequential tuning strategy identifies the best configuration as **dropout** $p = 0.2$ **+ weight decay** 10^{-4} **with learning rate** 3×10^{-4}, achieving peak validation accuracy **0.8982** (validation error **0.1018**) at epoch 18 on a fixed validation split. Most of the gain comes from dropout, with weight decay providing a smaller additional improvement and learning-rate tuning producing the final uplift. Batch Normalisation is competitive but does not exceed the selected configuration, and an augmentation + label smoothing attempt underperforms and is treated as a negative result (noting it used a different training/evaluation schedule). Overall, simple regularisation and careful optimisation are sufficient to substantially reduce the baseline's generalisation gap for this MLP-based benchmark.

References

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